

CompTIA A+ Core 1 Playbook

IT Labs

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Install a Power Supply

You are the IT administrator for a small corporate network. You are repairing the computer in the IT Administration Office, which appears to have a failed power supply. After testing the power supply and confirming the failure, you removed it from the computer. Now, you need to select a replacement power supply.

In this lab, your task is to:

- Install a power supply:
 - The power supply must have the appropriate power connectors for the motherboard and the CPU. Consult the motherboard documentation to see its power requirements.
 - The power supply must support adding a graphics card that requires its own power connector.
 - In the sim, the power supply is seated just below the motherboard.
- Make the following connections from the power supply:
 - Connect the motherboard power connector.
 - Connect the CPU power connector.
 - Connect the power connectors for the SATA hard drives.
- Start the computer:
 - Plug the computer in using the existing cable plugged into the power strip.
 - Turn on the power supply using the switch on the back.
 - Turn on the computer and boot into Windows.

Complete this lab as follows:

1. Locate the correct power supply to install.
 1. Switch the computer to the **Motherboard** view and select the motherboard.
 2. From the Selected Item pane, select the **Item Detail** icon to view the documentation.
 3. On the list of ports, #2 shows that the motherboard requires 24-pin and 8-pin connectors.
 4. Close the Item Detail window.
 5. Under Inventory, expand **Power Supplies**.
 6. Select a **power supply**.
 7. From the Selected Item pane, examine each of the **power connectors** (look for all the connectors required for this scenario).
 8. Repeat steps 1f–1g for each power supply.
2. Install the correct power supply.
 1. Under Inventory, drag the correct **power supply** to the area in the case below the motherboard.
 2. From the Selected Item pane, drag the **20+4 pin connector** to the motherboard to connect the motherboard's main power connector, which is on the right edge of the motherboard on the top half.
 3. From the Selected Item pane, drag the **Power Supply, 8-pin CPU** connector to the motherboard to connect the CPU power connector, which is on the top edge of the motherboard on the left side.
3. Connect the SATA hard drives.
 1. From the Selected Item pane, drag a **SATA, 15-pin** power connector to the power port on the top hard drive.
 2. From the Selected Item pane, drag a different **SATA, 15-pin** power connector to the power port on the second hard drive.
 3. From the Selected Item pane, drag a different **SATA, 15-pin** power connector to the power port on the third hard drive.
4. Plug the computer into the wall outlet.
 1. Switch to the **Back** view of the computer.
 2. Select the top **power cord** from the power strip.
 3. From the Selected Item pane, drag the **Connector, Power, AC, Female** connector to the power supply port on the back of the computer.
 4. On the power supply, select the **power switch** to move it to the On position.

5. Above the computer, select **Front**.
6. On the computer, select the **power** button to turn the computer on and boot into Windows.